Week 2

Spatial Data Exploration

Course zoom link for Summer Quarter 2025

July 8, 2025

*Note that this course will be recorded🎥

2025 LA Wildfires: Recovery Resources

Volunteer, recovery, and relief resources

- Sign up for BruinALERT from Office of Emergency Management for UCLA Emergency and Evacuation updates.
- Mutual Aid LA Google Doc resources
- LAist: Want to help fire victims? Here's what experts say does the most good and places seeking volunteers
- LAist: A quick roundup of SoCal fire coverage and safety resources
- LAist: Help for fire victims

GIS resources

- Watch Duty Live Wildfire Maps and Evacuation Notifications
- Air Quality map

Project formation for midterm and final projects

- Open and review the midterm and final project guidelines.
- The form is not graded, but it will help you with your midterm proposal due before Week 3. You are welcome to submit the project proposal assignment when your group is ready. It is due July 7th at 11:59 pm.

Hands on Lab

First, grab the course material, and "pull" it into your JupyterHub space:

- Git related

Note that you have to do this at the start of every lecture to get the latest material.

Assignments (due 11:59 pm, Monday, July 14th the day before class)

Individual assignment: Data Exploration

Create a token

In order to pull and push content to GitHub, you must first create a token, which will serve as your password. Refer to this tutorial to create your token:

- How to create a token

Clone your repo

This is your first code assignment submission. Before you begin, create a clone of your repo in JupyterHub.

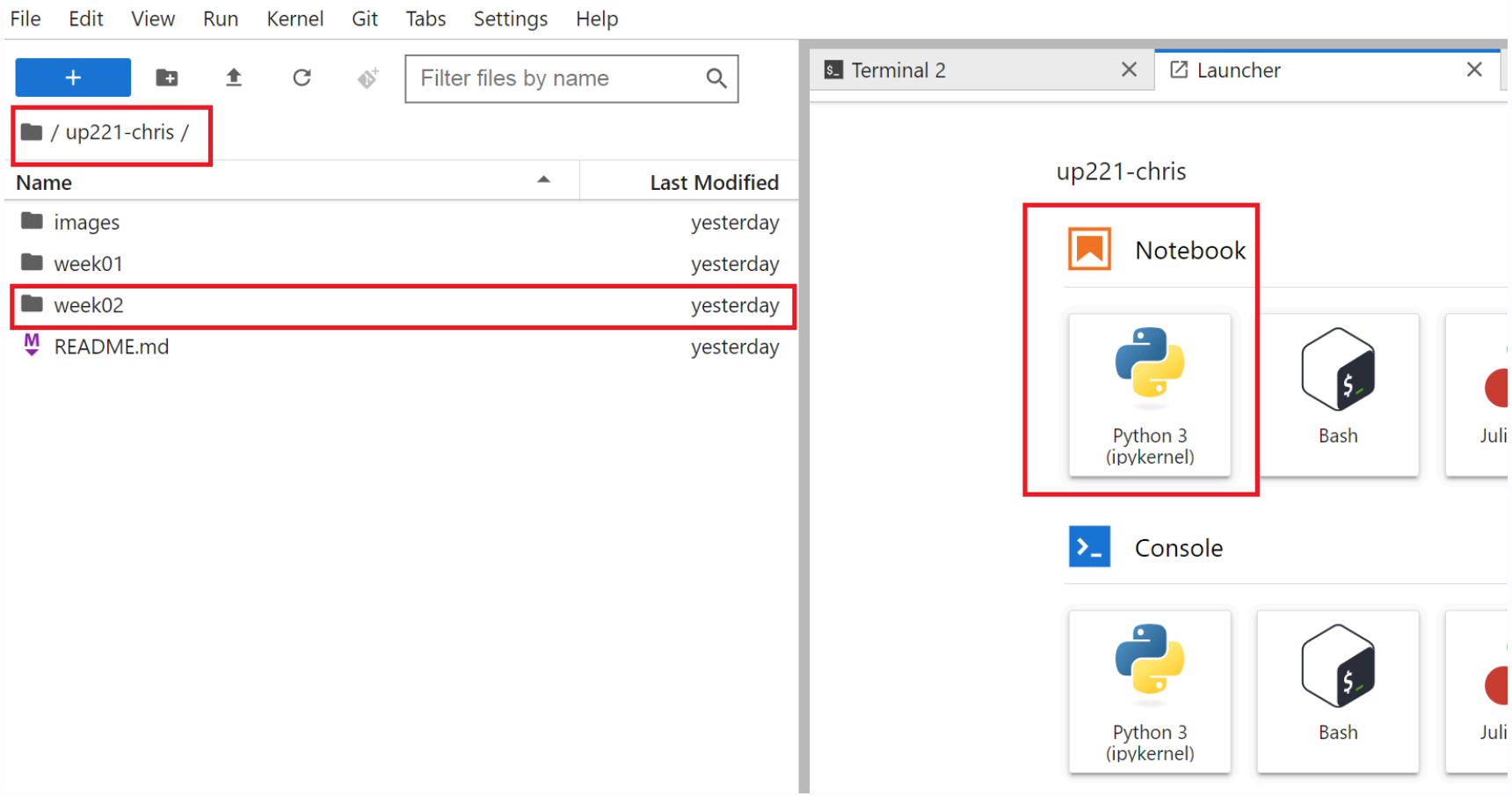
- How to clone your repo into JupyterHub

Submission guidelines:

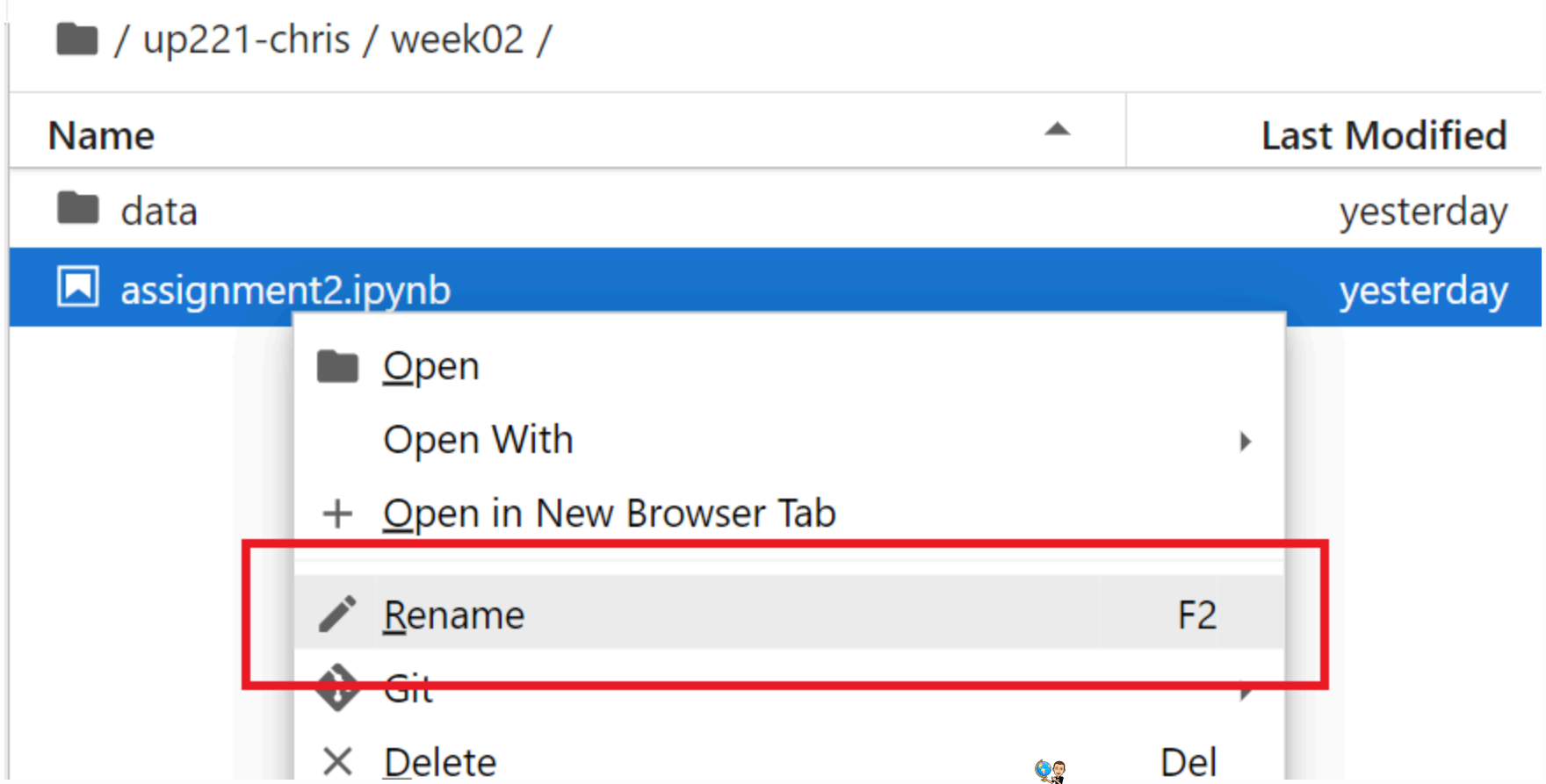
- Find and download a dataset of your choice. This can be a shapefile, csv file, or json file. For many of you, you may have already done this as part of your week 1 assignments.
- Launch JupyterHub, go to your `pa405` repo folder, and create a `week02` folder.
- Load the dataset to the `pa405/week02` folder.

Create a new python notebook

Do not work on a copy of the lab notebook



Right click on the `Untitled.ipynb` tab and rename the notebook to `week2assignment.ipynb` or `assignment2.ipynb`. Just be consistent throughout the quarter.



Add an introductory markdown cell with a title (header) and paragraph that describes what you are doing.

Import the data, and conduct data exploration, making sure to document your steps and your preliminary findings. At minimum, run the following commands:

- `.shape`
- `.info`
- `.head()`
- `.plot()`
- `.value_counts()`
- run a query on the data that filters it in some way

For each code cell, add a markdown cell that explains what you are doing.

Add markdown cells that describe the output of each operation.

Save your notebook.

Commit your changes to your GitHub class repo.

Commit your changes to your GitHub repo by following these instructions:

- How to commit and push to your repo

Submit your assignment

The last step is to submit your assignment to the class repo discussion section here.

Project Assignment #1

Project Proposal

The course will largely be guided by your final project, which will be conducted individually. Consider that your final project will be a representation of what you learn in this course, and how you apply it on a project level. Consider also that your final project can serve to showcase your data science and mapping skills, which may become a valuable asset for your career moving forward.

Step 1: Plan

Discuss and identify a research question, identify and collect data, articulate how you intend to use and analyze it, and begin to speculate how the data can answer your research inquiry. Understand that this may change later, especially as we learn more about what we can do with our methods, and also, as you find the data sources that can or cannot support your research.

Step 2: Data

Next, identify and download at least two datasets that you believe can guide your research. Make sure that at least one of them has a spatial component. For example, if one of your datasets comes from the census, identify which survey you will use (e.g. Decennial or American Community Survey), and which variables you will select. Note that we will be covering census data in detail in Week 3. Your second dataset should come from a separate source. For example, you may want to do a crime analysis and obtain data from the LA Data Portal.

Step 3: Create a project repo.

1. Create a new repository for your project
2. Go to Settings, Manage Access, and add your partner as a collaborator
3. Create a Project Assignments folder (hint: click on Add File, Create new File, and enter "Project Assignments/readme.md" in the text box). I recommend that you do this in Jupyter Lab.
4. Create a markdown file for your project proposal

Step 4: Write your proposal

Your project proposal should include the following:

- An introduction of your research question
- An explanation of why it is important to you, why it matters to others, and what is at stake
- A description of the spatial scope (e.g. Boyle Heights or Hong Kong), and why space and/or time matters for your project
- A preliminary but definitive description of data sources (at least two) that you will use
  - Include datasource with links
- A scope that explains the intended analysis and resulting visualizations for your project
- A concluding paragraph of what insights you expect to gain from your research

Step 5: Submit

Submit your assignment here